

Factorization Exercises — Solutions

1 1 Basic Level Solutions

1. $3x + 6 = 3(x + 2)$
2. $5a^2 - 10a = 5a(a - 2)$
3. $x^2 - 9 = (x + 3)(x - 3)$
4. $4y^2 - 25 = (2y + 5)(2y - 5)$
5. $6m^2 + 9m = 3m(2m + 3)$

2 2 Intermediate Level Solutions

1. $x^2 + 6x + 9 = (x + 3)^2$
2. $a^2 - 8a + 16 = (a - 4)^2$
3. $x^2 + 5x + 6 = (x + 2)(x + 3)$
4. $x^2 - 2x - 8 = (x - 4)(x + 2)$
5. $2x^2 + 7x + 3 = (2x + 1)(x + 3)$

3 3 Advanced Level Solutions

1. $ax + ay + bx + by = (a + b)(x + y)$
2. $x^3 - 4x = x(x^2 - 4) = x(x + 2)(x - 2)$
3. $x^2 - y^2 + x + y = (x + y)(x - y) + (x + y) = (x + y)(x - y + 1)$
4. $3x^2 - 12xy + 12y^2 = 3(x^2 - 4xy + 4y^2) = 3(x - 2y)^2$
5. $x^4 - 16 = (x^2 + 4)(x^2 - 4) = (x^2 + 4)(x + 2)(x - 2)$
6. $x^3 + 3x^2 - 4x - 12 = x^2(x + 3) - 4(x + 3) = (x^2 - 4)(x + 3) = (x + 2)(x - 2)(x + 3)$